

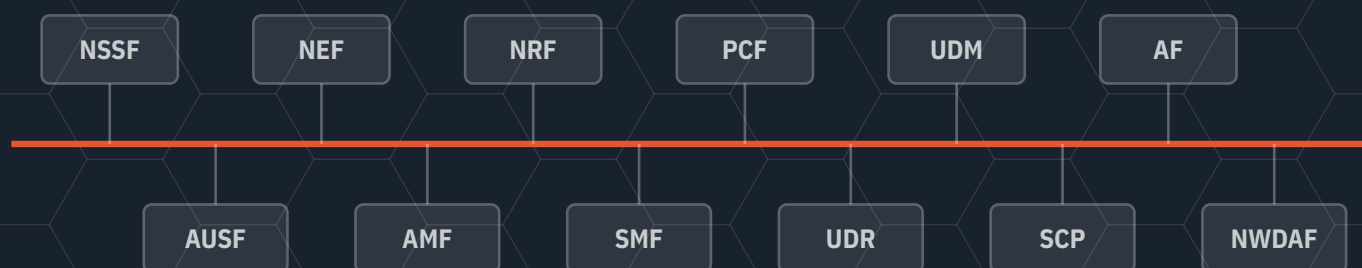
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 18

5G Service-Based Architecture (SBA)

معماری خدمات‌محور 5G (SBA)



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari

In This Module

1	Why 5G Moved Away from EPC Point-to-Point Interfaces	3
2	SBA Design Principles	4
3	5G Core Network Functions	6
4	NRF — The Key Enabler of SBA	10
5	Service Discovery and Communication Flow	13
6	SBI Protocol Stack	15
7	Comparison: EPC Interfaces vs 5G SBA	16
	Summary	18
	Review Questions	18
	References	18

Learning Objectives

By the end of this module, you will be able to:

- Explain why 5G abandoned EPC's point-to-point interface model
- Describe SBA design principles and the role of HTTP/2, JSON, and REST
- Identify all 5G Core Network Functions and their services
- Explain the NRF's critical role in service discovery
- Compare direct vs indirect (SCP) communication models
- Contrast EPC architecture with 5G SBA

1 Why 5G Moved Away from EPC Point-to-Point Interfaces

1.1 The EPC Problem

In 4G/LTE, the Evolved Packet Core (EPC) uses **point-to-point reference points** between network elements:

Element Pair	Interface
MME ↔ S-GW	S11
MME ↔ HSS	S6a
S-GW ↔ P-GW	S5/S8
MME ↔ eNB	S1-MME
P-GW ↔ PCRF	Gx

The scaling problem:

- With N network elements, you need up to $N \times (N-1)/2$ unique reference points
- Each interface has its own protocol specification (Diameter, GTP-C, etc.)
- Adding a new network element requires defining new interfaces to every existing element
- Rigid coupling makes evolution extremely difficult
- Vendor interoperability requires testing every interface combination

Example: EPC with 6 core elements → up to 15 unique interfaces, each with its own protocol, message formats, and procedures.

1.2 The SBA Solution

5G introduces **Service-Based Architecture** where:

- Network Functions (NFs) expose **services** via a common interface framework
- Any NF can **discover and consume** any other NF's services
- Adding a new NF only requires registering its services — no new interfaces needed
- Common protocol stack (HTTP/2 + JSON) for all control plane communication

1.3 Cloud-Native Principles

SBA embraces cloud-native design:

Principle	Implementation in 5G SBA
Microservices	Each NF is decomposed into independent services
Stateless	NF instances don't store session state locally (externalized to data stores)
Horizontal scaling	Add more NF instances behind load balancers
Containerization	NFs deployed as containers in Kubernetes
CI/CD	Independent NF lifecycle management
Service mesh	SCP provides routing, load balancing, observability

2 SBA Design Principles

2.1 Service-Based Interfaces (SBI) vs Reference Points

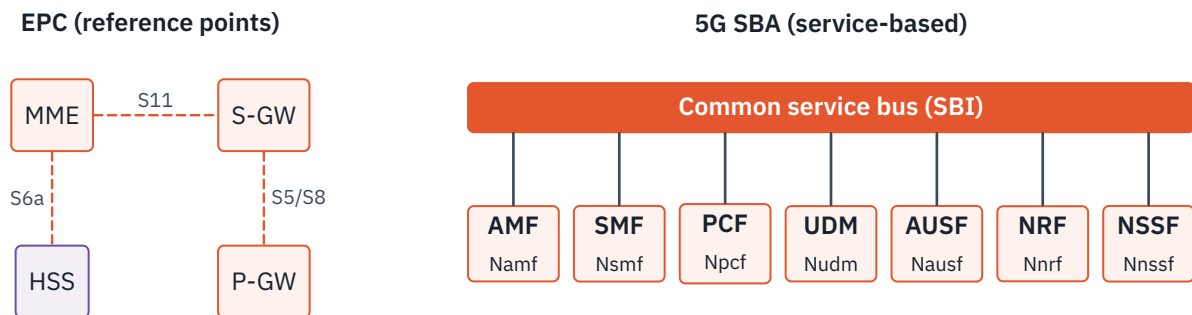


Figure 18.1 EPC reference points versus the 5G service-based bus

Reference Points (4G): Each line is a unique protocol/interface specification

Service-Based Interfaces (5G): All NFs connect to a common bus using the same protocol framework

2.2 NF Service Interaction Models

NFs communicate using two patterns:

1. Request-Response (Synchronous)

- NF Consumer sends a request to NF Producer
- Producer processes and returns a response
- Example: AMF queries UDM for subscriber data

2. Subscribe-Notify (Asynchronous)

- NF Consumer subscribes to events from NF Producer
- Producer sends notifications when events occur

- Example: AMF subscribes to UDM for subscription data changes

2.3 HTTP/2 as Transport Protocol

5G SBA chose HTTP/2 over HTTP/1.1 for critical reasons:

Feature	Benefit for 5G
Multiplexing	Multiple requests/responses over single TCP connection — reduces latency
Header compression (HPACK)	Reduces overhead for repetitive signaling headers
Server push	Enables efficient notification delivery
Binary framing	More efficient parsing than text-based HTTP/1.1
Stream prioritization	Critical signaling (e.g., emergency calls) gets priority

2.4 JSON as Data Format

Reason	Explanation
Human-readable	Easy debugging, logging, troubleshooting
Web-native	Leverages massive web ecosystem tooling
Tooling	Validators, editors, parsers available in every language
Schema support	OpenAPI/JSON Schema for API definition
Lightweight	Less verbose than XML (used in older 3GPP specs)

2.5 REST API Design Principles

5G SBA APIs follow RESTful conventions:

- **Resources** identified by URIs: <https://{nfInstanceId}.5gc.mnc{MNC}.mcc{MCC}/nsmf-pdusession/v1/sm-contexts>
- **HTTP methods** map to operations: GET (read), POST (create), PUT (update), DELETE (remove), PATCH (partial update)
- **Stateless** requests — each request contains all necessary context
- **HATEOAS** — responses include links to related resources
- **API versioning** — version in URI path (v1, v2)

3 5G Core Network Functions

■ SBA Architecture Diagram

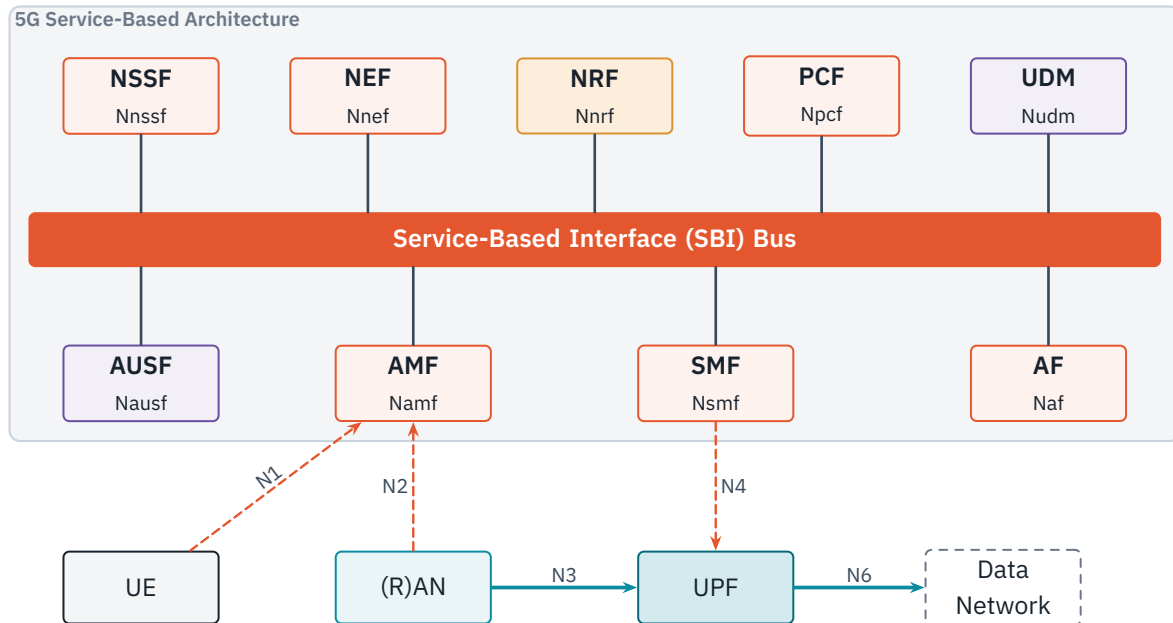


Figure 18.2 5G service-based architecture around the SBI

3.1 AMF – Access and Mobility Management Function

Purpose: Handles all access and mobility-related signaling for the UE.

Service Interface: [Namf](#)

Key Services:

Service	Description
Namf_Communication	N1/N2 message transport, UE-specific signaling
Namf_EventExposure	Expose mobility events (location, reachability)
Namf_MT	Mobile-terminated message delivery
Namf_Location	UE location services

Key Responsibilities:

- UE registration and deregistration
- Connection management (idle ↔ connected)
- Mobility management (handover, tracking area updates)
- Security context management (NAS encryption/integrity)
- Relay NAS-SM messages between UE and SMF

3.2 SMF – Session Management Function

Purpose: Manages PDU (data) sessions between UE and Data Network.

Service Interface: `Nsmf`

Key Services:

Service	Description
<code>Nsmf_PDUSession</code>	Create, modify, release PDU sessions
<code>Nsmf_EventExposure</code>	PDU session events (e.g., IP address change)

Key Responsibilities:

- PDU session establishment, modification, release
- UE IP address allocation and management
- UPF selection and control (via N4/PFCP)
- QoS flow management and enforcement
- Downlink data notification
- Traffic steering and routing configuration

3.3 UPF – User Plane Function

Purpose: Forwards user data packets between RAN and Data Networks.

⚠ **Note:** UPF does NOT use SBI. It uses reference points:

- **N3** – between RAN and UPF (GTP-U tunnel)
- **N4** – between SMF and UPF (PFCP protocol)
- **N6** – between UPF and Data Network (IP traffic)
- **N9** – between UPFs (inter-UPF GTP-U tunnel)

Key Responsibilities:

- Packet routing and forwarding
- QoS handling (marking, policing, shaping)
- Traffic usage reporting
- Uplink traffic verification
- Lawful intercept (user plane)
- Buffering of downlink packets (UE in idle mode)

3.4 UDM – Unified Data Management

Purpose: Manages subscriber data and computes authentication credentials.

Service Interface: `Nudm`

Key Services:

Service	Description
<code>Nudm_SubscriberDataManagement</code>	Retrieve/update subscription data

[Nudm_UEContextManagement](#)	Track which AMF/SMF serves a UE
[Nudm_UEAuthentication](#)	Generate authentication vectors
[Nudm_EventExposure](#)	Subscription data change notifications

Key Responsibilities:

- Stores subscription data (authorized slices, QoS profiles, DNN permissions)
- Computes authentication credentials (using SUPI, K, OPc)
- Manages UE context registration (serving AMF, serving SMF)
- De-concealment of SUCI → SUPI

3.5 AUSF – Authentication Server Function

Purpose: Executes the authentication procedure for UEs.

Service Interface: [Nausf](#)

Key Services:

Service	Description
—	—
[Nausf_UEAuthentication](#)	Perform 5G-AKA or EAP-AKA' authentication
[Nausf_SoRProtection](#)	Steering of Roaming security

Key Responsibilities:

- Authenticates UEs during registration
- Supports 5G-AKA and EAP-AKA' methods
- Stores intermediate key material (K_{AUSF})
- Interfaces with UDM for authentication vectors

3.6 PCF – Policy Control Function

Purpose: Provides policy rules governing network behavior and QoS.

Service Interface: [Npcf](#)

Key Services:

Service	Description
—	—
[Npcf_SMPolicyControl](#)	Session-level policies (QoS, charging, gating)
[Npcf_AMPolicyControl](#)	Access and mobility policies (RFSP, service area)
[Npcf_PolicyAuthorization](#)	AF-triggered policy (e.g., IMS requests QoS)
[Npcf_BDTPolicyControl](#)	Background data transfer policies
[Npcf_EventExposure](#)	Policy-related event notifications

Key Responsibilities:

- QoS policy decisions
- Charging policy rules
- Access control and gating decisions
- Network slicing policies

- Usage monitoring control

3.7 NRF — Network Repository Function

Purpose: The **central registry** of all NF instances and their services — the backbone of SBA.

Service Interface: `Nnrf`

Key Services:

Service	Description
<code>Nnrf_NFManagement</code>	NF registration, update, deregistration
<code>Nnrf_NFDiscovery</code>	Query available NFs by type, slice, location
<code>Nnrf_AccessToken</code>	Issue OAuth2.0 tokens for NF-to-NF authorization

—|—|

★ **NRF is the critical enabler of SBA** — without it, NFs cannot discover each other and the service-based model collapses back to static configuration.

3.8 NSSF — Network Slice Selection Function

Purpose: Selects the appropriate network slice instance for a UE.

Service Interface: `Nnssf`

Key Services:

Service	Description
<code>Nnssf_NSSelection</code>	Determine allowed slices and target AMF set
<code>Nnssf_NSSAIAvailability</code>	Manage NSSAI availability per tracking area

—|—|

Key Responsibilities:

- Determines the Allowed NSSAI for UE registration
- Selects target AMF Set/AMF based on requested slices
- Manages per-TA NSSAI availability information

3.9 NEF — Network Exposure Function

Purpose: Securely exposes 5G network capabilities to external applications.

Service Interface: `Nnef`

Key Services:

Service	Description
<code>Nnef_EventExposure</code>	Expose network events to AFs
<code>Nnef_PFDManagement</code>	Packet Flow Description management
<code>Nnef_TrafficInfluence</code>	Allow AF to influence traffic routing
<code>Nnef_ParameterProvision</code>	Provision expected UE behavior

—|—|

Key Responsibilities:

- API gateway between 5G core and external world
- Translates internal 5GC information to external representations
- Enforces security policies on external access
- Rate limiting and API management

3.10 AF — Application Function

Purpose: Represents third-party applications that interact with the 5G core.

Service Interface: `Naf`

Key Responsibilities:

- Requests QoS for application flows (e.g., video streaming)
- Influences traffic routing (e.g., edge computing)
- Subscribes to network events (e.g., UE location)
- Can access 5GC directly (trusted) or via NEF (untrusted)

Examples: IMS (VoNR), streaming platforms, enterprise applications, MEC orchestrators.

4 NRF — The Key Enabler of SBA

The NRF is the **most critical component** that makes SBA work. Without NRF, 5G would revert to static, pre-configured connections like EPC.

4.1 NF Registration

When an NF instance starts, it **registers** with the NRF by providing its **NF Profile**:

```

JSON
{
  "nfInstanceId": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
  "nfType": "SMF",
  "nfStatus": "REGISTERED",
  "heartBeatTimer": 30,
  "plmnList": [{ "mcc": "310", "mnc": "014" }],
  "sNssai": [{ "sst": 1, "sd": "000001" }],
  "fqdn": "smf1.5gc.mnc014.mcc310.3gppnetwork.org",
  "ipv4Addresses": ["10.10.10.5"],
  "nfServices": [
    {
      "serviceInstanceId": "nsmf-pdusession-01",
      "serviceName": "nsmf-pdusession",
      "versions": [{ "apiVersionInUri": "v1", "apiFullVersion": "1.2.0" }],
      "scheme": "https",
      "nfServiceStatus": "REGISTERED",
      "capacity": 100,
      "load": 25,
      "priority": 1
    }
  ],
  "smfInfo": {
    "sNssaiSmfInfoList": [{ "sNssai": { "sst": 1 }, "dnnSmfInfoList": [{ "dnn":
      ↪ "internet" }] } ]
  },
  "capacity": 100,
  "load": 25,
  "locality": "DC-East-1"
}

```

NF Profile contains:

- **NF Type** — what kind of NF (AMF, SMF, PCF, etc.)
- **Services offered** — list of service names and versions
- **Capacity** — relative capacity for load balancing (0-65535)
- **Load** — current load percentage (0-100)
- **Locality** — geographic/data center location for proximity-based selection
- **PLMN** — which networks this NF serves
- **S-NSSAI** — which slices this NF supports
- **IP/FQDN** — how to reach this NF

4.2 NF Discovery

When an NF needs a service, it queries the NRF:

```

GET /nnrf-disc/v1/nf-instances?target-nf-type=SMF&requester-nf-type=AMF&snssai={ "sst":1, "sd": "000001" }&dnn=internet

```

Discovery query parameters:

| Parameter | Purpose |

|—|—|

| **target-nf-type** | Type of NF being searched (mandatory) |

| **requester-nf-type** | Type of the requesting NF (mandatory) |

| **service-names** | Specific services needed |

| **snssai** | Network slice filter |

| **dnn** | Data Network Name filter |

| **tai** | Tracking area for locality |

| **preferred-locality** | Prefer NFs in same data center |

NRF returns: Ranked list of matching NF instances with endpoints and load information.

4.3 NF Status Notification

NFs can **subscribe** to NRF for status events:

- NF registered / deregistered
- NF profile changed (load, capacity)
- NF service status changed

This enables:

- **AMF** to be notified when new SMFs become available
- **Load balancers** to update routing when NF load changes
- **Fault detection** when NF heartbeat expires

NF Discovery Flow Diagram

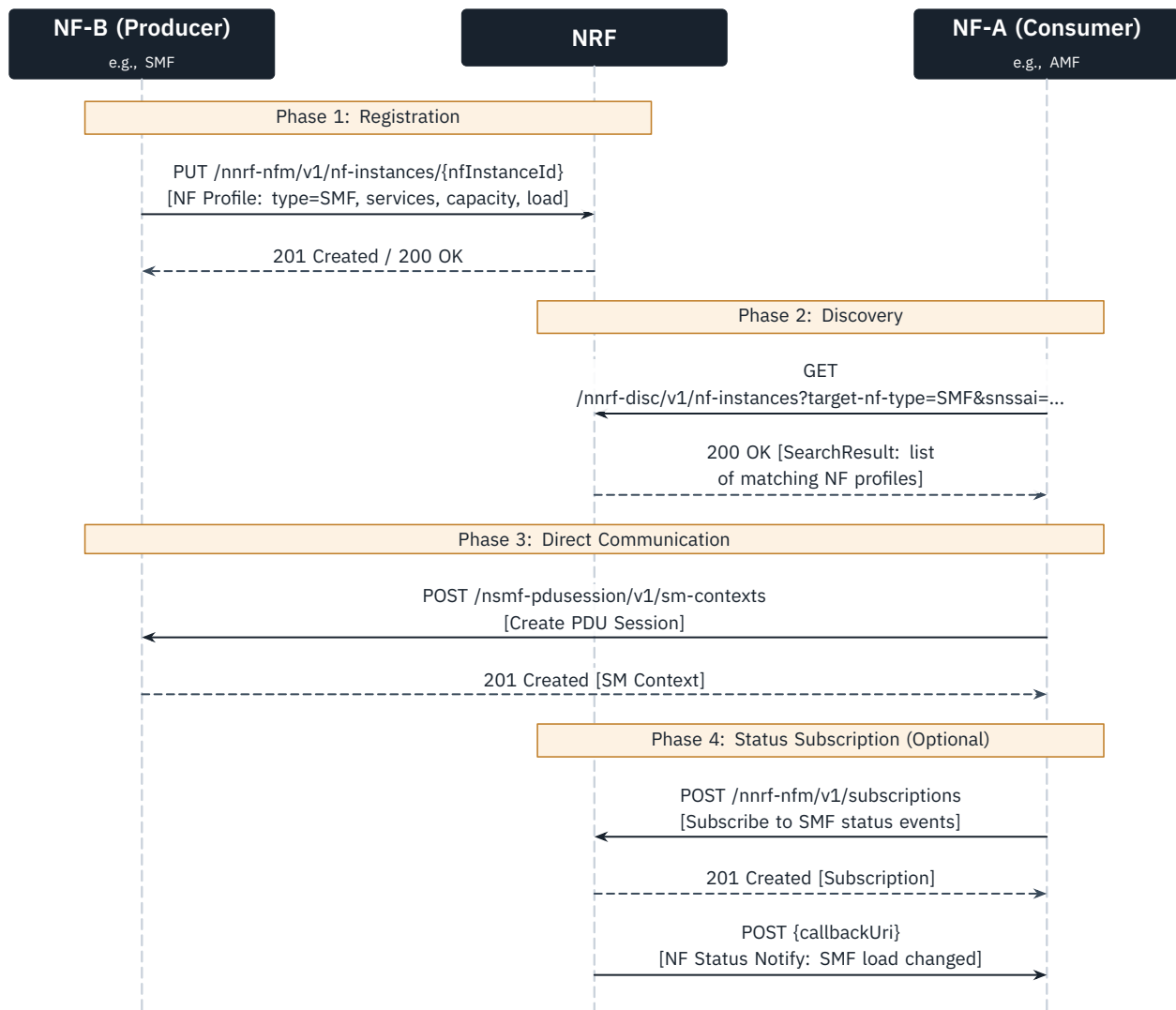


Figure 18.3 NF registration, discovery and subscription through the NRF

5 Service Discovery and Communication Flow

5.1 Direct Communication

In the **direct model**, NF-A communicates directly with NF-B after discovery:

1. NF-A → NRF: "Find me an SMF for slice 1, DNN internet"
2. NRF → NF-A: "Here are 3 SMFs: SMF-1 (load=20%), SMF-2 (load=45%), SMF-3 (load=80%)"
3. NF-A selects SMF-1 (lowest load)
4. NF-A → SMF-1: Direct service request

Advantages: Low latency (no intermediate hop), simple

Disadvantages: NF-A must handle load balancing, retries, failover logic

5.2 Indirect Communication via SCP

The **Service Communication Proxy (SCP)** acts as an intermediary:

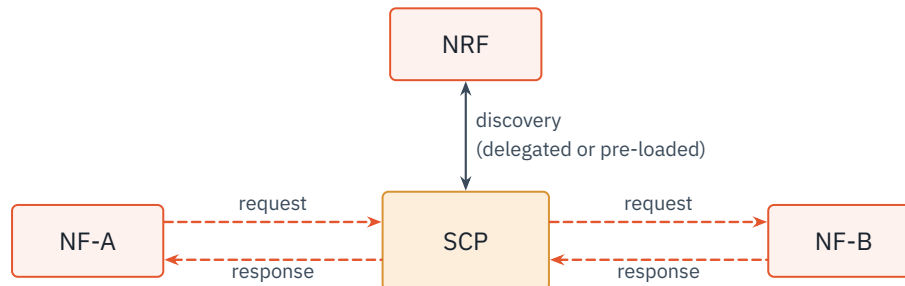


Figure 18.4 Indirect communication through the SCP

SCP provides:

- Delegated discovery (SCP queries NRF on behalf of NF-A)
- Load balancing across NF-B instances
- Message routing and topology hiding
- Retry and failover logic
- Monitoring and observability
- Connection pooling

Communication Models (3GPP TS 23.501):

| Model | Discovery | Routing |

|—|—|—|

| Model A | NF-A via NRF | Direct NF-A → NF-B |

| Model B | NF-A via NRF | Via SCP (NF-A → SCP → NF-B) |

| Model C | Delegated to SCP | Via SCP (NF-A → SCP → NF-B) |

| Model D | Delegated to SCP | Direct NF-A → NF-B |

6 SBI Protocol Stack

Complete Protocol Stack

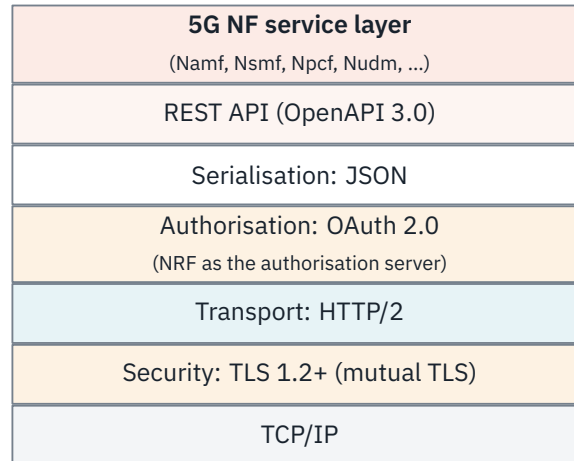


Figure 18.5 SBI protocol stack

OAuth 2.0 for NF Authorization

Before NF-A can call NF-B's service, it must obtain an **access token** from the NRF:

1. NF-A → NRF: POST /oauth2/token

```
{
  "grant_type": "client_credentials",
  "nfInstanceId": "{NF-A ID}",
  "nfType": "AMF",
  "targetNfType": "SMF",
  "scope": "nsmf-pdusession"
}
```
2. NRF → NF-A: Access Token (JWT)

```
{
  "access_token": "eyJ...",
  "token_type": "Bearer",
  "expires_in": 3600,
  "scope": "nsmf-pdusession"
}
```
3. NF-A → NF-B: Service Request + Authorization: Bearer eyJ...
4. NF-B validates the token (signature, expiry, scope)

7 Comparison: EPC Interfaces vs 5G SBA

I EPC vs SBA Comparison Diagram

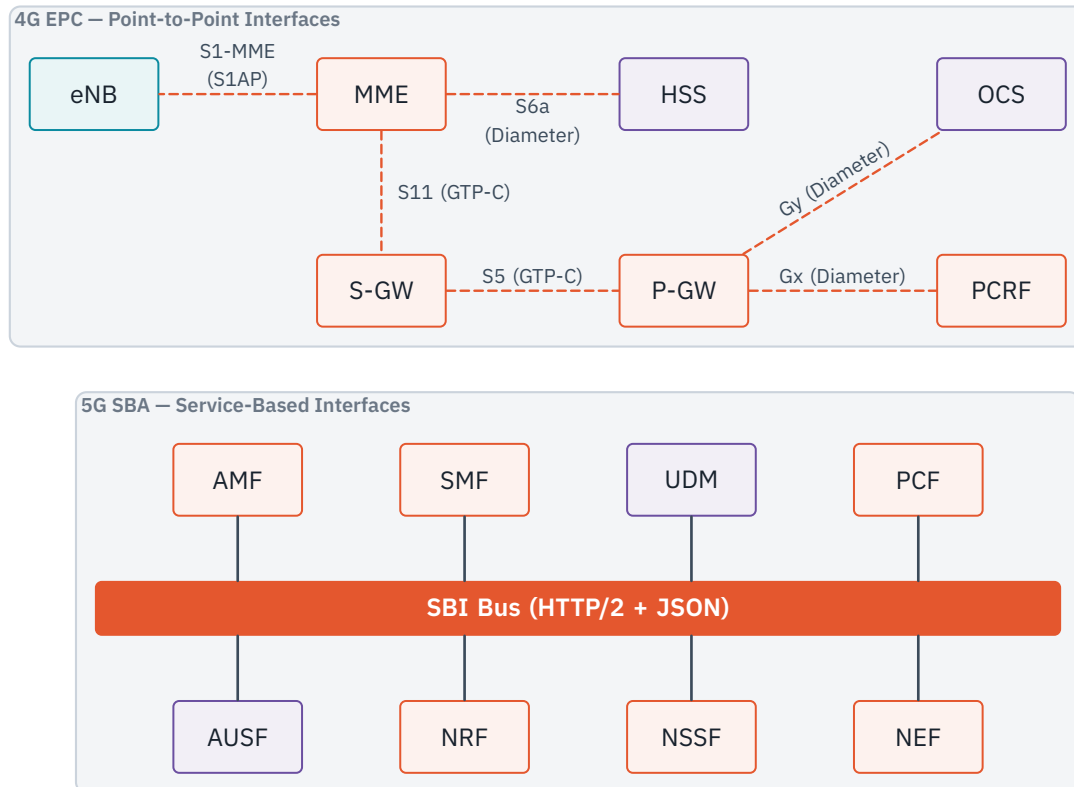


Figure 18.6 Point-to-point EPC interfaces versus the SBA bus

Detailed Comparison Table

Aspect	4G EPC	5G SBA
Architecture	Point-to-point reference points	Service-based interfaces on common bus
Protocols	Diameter, GTP-C, S1AP (multiple)	HTTP/2 + JSON (unified)
Coupling	Tight — each pair has unique interface	Loose — common framework for all
Adding new NF	Define new interfaces to all peers	Register services with NRF
Discovery	Static configuration	Dynamic via NRF
Scaling	Scale entire monolithic element	Scale individual NF services independently
Deployment	Hardware appliances / VMs	Cloud-native containers (Kubernetes)
State	Stateful network elements	Stateless NFs (externalized state)
Redundancy	Active-standby pairs	Horizontal scaling + NRF re-selection
Load balancing	DNS-based or static	NRF-based (capacity, load, locality)
Authorization	Hop-by-hop TLS	End-to-end OAuth 2.0 tokens
API definition	3GPP-specific ASN.1/XML	OpenAPI 3.0 (industry standard)
Extensibility	New release = new interfaces	New services registered dynamically
Vendor ecosystem	Telecom-specific vendors	Cloud/IT vendors can participate

Functional Mapping: EPC → 5G

EPC Element	5G Equivalent	Key Difference
MME	AMF + SMF	Separated access/mobility from session mgmt
S-GW + P-GW	UPF	Simplified to single user plane element
HSS	UDM + AUSF	Separated data storage from authentication
PCRF	PCF	Same role, now on SBI
— (new)	NRF	Service registry (no EPC equivalent!)
— (new)	NSSF	Slice selection (no slicing in 4G)
— (new)	NEF	Replaces SCEF with richer API exposure

Summary

Concept	Key Takeaway
SBA motivation	EPC's $N \times (N-1)/2$ interfaces don't scale; SBA uses a common bus
Design principles	HTTP/2, JSON, REST, microservices, stateless
NRF	Heart of SBA — registration, discovery, authorization
Communication	Direct (NF→NF) or Indirect (via SCP)
Security	mTLS + OAuth 2.0 tokens from NRF
Key difference	EPC = static, rigid, monolithic; SBA = dynamic, flexible, cloud-native

Review Questions

1. Calculate how many unique interfaces are needed for 8 EPC elements. How does SBA solve this?
2. Why was HTTP/2 chosen over HTTP/1.1 for SBI?
3. What information is included in an NF Profile during NRF registration?
4. Why is the UPF not connected to the SBI bus?
5. Compare SCP communication Models B and C — when would you use each?
6. How does OAuth 2.0 work in the 5G SBA context? What role does NRF play?
7. An operator wants to add a new Analytics NF (NWDAF). Compare the effort in EPC vs SBA architecture.
8. Why is NRF considered the “single most critical” component of SBA?

References

- 3GPP TS 23.501 — System Architecture for the 5G System (Stage 2)
- 3GPP TS 23.502 — Procedures for the 5G System (Stage 2)
- 3GPP TS 29.510 — NRF Services (Stage 3)
- 3GPP TS 29.500 — Technical Realization of Service-Based Architecture
- 3GPP TS 33.501 — Security Architecture and Procedures for 5G System